A SURJECTION FROM THE MODULE TO A FUSION PRODUCT AND THE RESULTING LOWER BOUND FOR ITS DIMENSION

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Allowed standard prerequisites (editorial).

- The fusion product of graded modules with its notation, and the modules D(1, theta) and the evaluation module of the fundamental representation appearing as its factors. The author's sections 3 and 4.1, where the modules and the fusion product are set up before the counted statement. Fixes the target of the asserted surjection and the two factors whose dimensions occur in the inequality.
- The defining relations of the cyclic module on the source side, and the annihilation of the generators of the two factors by twisted positive and negative root vectors. Used inside the counted proof (source lines 670-697). Supplies the relations which, by the lemma labelled f2, produce the annihilation needed to verify the presentation.
- The statement that annihilation in the factors passes to the fusion product. Invoked inside the counted proof at source line 686 as a pointer to the lemma labelled f2 (a navigation-pointer stub). Transfers the relation satisfied by the image of the generator to the fusion product.

Source reference. Original reference f2: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 686–686. Navigation pointer only: the counted text refers to the item f2 of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

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Proposition 1. *The map* $\psi : \mathbf{V}_{i,k+1} \to D(1,\theta)^{*(k+1-i)} * \mathrm{ev}_0 V(\theta)^{*i}$ *such that* $\psi(\overline{v}_{i,k+1}) = \overline{w}_\theta^{*(k+1-i)} * v_\theta^{*i}$ *is well-defined and surjective morphism of* $\mathfrak{g}[t]$*-modules. In particular,*

$$\dim \mathbf{V}_{i,k+1} \geq (\dim D(1,\theta))^{k+1-i} (\dim V(\theta))^i. \tag{1}$$

Proof. We only need to show that $\psi(\overline{v}_{i,k+1})$ satisfies the defining relations of $\mathbf{V}_{i,k+1}$. But they follow easily from the following relations:

$$((h \otimes 1) - \langle (k+1)\theta, h\rangle)\big(\overline{w}_\theta^{\otimes(k+1-i)} \otimes v_\theta^{\otimes i}\big) = 0, \qquad \forall\, h \in \mathfrak{h},$$
$$(x_\alpha^- \otimes 1)^{\langle (k+1)\theta,\ \ \alpha^\vee\rangle + 1}\big(\overline{w}_\theta^{\otimes(k+1-i)} \otimes v_\theta^{\otimes i}\big) = 0, \qquad \forall\, \alpha \in R^+$$

(which holds in $D(1,\theta)^{\otimes(k+1-i)} \otimes \mathrm{ev}_0 V(\theta)^{\otimes i}$) and $(h \otimes t^s)\psi(\overline{v}_{i,k+1}) = 0, \forall\, s \geq 1$, $h \in \mathfrak{h}$ (which holds in $D(1,\theta)^{*(k+1-i)} * \mathrm{ev}_0 V(\theta)^{*i}$). Further from Lemma f2, by using the relations

$$(x_\alpha^+ \otimes t^s)\overline{w}_\theta = 0 = (x_\alpha^+ \otimes t^s)v_\theta, \qquad \forall\, s \geq 0, \quad \alpha \in R^+,$$
$$(x_\alpha^- \otimes t)\overline{w}_\theta = \big(x_\theta^- \otimes t^2\big)\overline{w}_\theta = 0 = (x_\theta^- \otimes t)v_\theta = (x_\alpha^- \otimes t)v_\theta, \qquad \forall\, \alpha \in R^+ \setminus \{\theta\},$$

which holds in $D(1,\theta)$ and $\mathrm{ev}_0 V(\theta)$. □